

Assignment 1 - Lectures 1 and 2

Instructions:

- The assignments are individual. You may consult and discuss with your colleagues, but you need to provide independent answers.
- You may use Matlab/Python/Julia to solve the assignment. Many questions require you clearly to do so.
- Provide detailed answers, describing the steps you followed.
- Provide clear reports, typed, preferably using LaTeX.
- Submit the reports digitally on Brightspace as indicated on the lectures.
- Include your code, for reproducibility of your results, in your Brightspace submission on a zip file, or through a link on your report.
- The code will *only* be used to verify reproducibility in case of doubts. The grading will be performed based on the results described in the report.

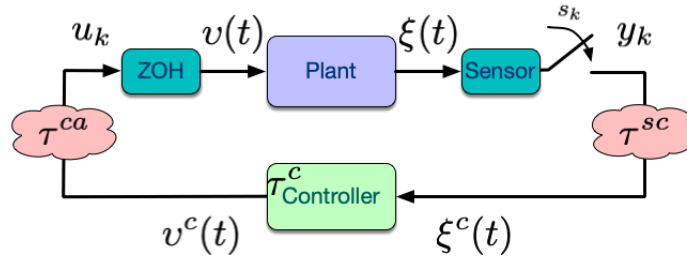


Figure 1: Schematic of a NCS including delays.

Consider a sample-data networked control system (NCS), as depicted in Figure 1. Assume that the plant dynamics are linear time-invariant given by:

$$\dot{\xi}(t) = A\xi(t) + Bv(t),$$

where the control signal v is a piece-wise constant signal resulting from the application of a controller in sampled-and-hold fashion, i.e. $v(t) = u_k, t \in [s_k, s_{k+1})$.

The system matrices are given by:

$$A = \begin{bmatrix} 0 & 0.5 - c \\ 0.2 + a - b & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

where a is given by the first digit, b by the third digit, and c by the last digit of your student ID number.

We start considering the system with a constant sampling interval $h = s_{k+1} - s_k$ for all k , and assuming no delay is present $\tau = \tau^{sc} + \tau^c + \tau^{ca} = 0$.

Important note: For all the questions that follow, you do **NOT** necessarily need to provide analytic expressions, unless explicitly stated. However, you need to describe the steps you took to produce the resulting plots, and design process you followed.

Question 1: (2p)

1. (2p) Design a linear continuous time controller $v(t) = -\bar{K}\xi(t)$ placing the poles of the continuous closed-loop at $-2 + j$ and $-2 - j$. Construct the exact discrete-time model $x_{k+1} = F(h)x_k + G(h)u_k$, and the closed-loop system resulting from applying the controller in sample-data (piece-wise constant) fashion: $u_k = -\bar{K}x_k$, $x_k := \xi(s_k)$. Provide ranges of sampling times for which the sampled-data system is stable. Comment on what you observe.

Next we consider the case when delays are present in the networked system. Assume that the system is affected by a **constant** small delay $\tau \in [0, h)$, and it is controlled with the *static* controller $\bar{K}$.

Question 2: (4p)

1. (2p) Construct the exact discrete-time model for the NCS with delays and study the combinations of sampling intervals and system delays that result in an asymptotically stable closed-loop. Provide plots illustrating the combinations of (h, τ) retaining stability.
2. (2p) Select a sampling interval h that guarantees stability under no delays. Redesign the controller to improve the robustness against delays, that is, try to design a controller that increases the range of tolerable delays for the selected h . Think about designing now a *dynamic* controller instead.

Next we consider a different approach to deal with delays $\tau \in [0, h)$. Design a second controller $\bar{K}_2$ such that the poles of the continuous closed-loop are placed at -1 and -3 . Now, assume that the controller applied is such that at time $t = s_{k-1} + \tau \in [s_{k-1}, s_k)$ the applied input is $u_{k-1} = -\bar{K}x(s_{k-1}) - \bar{K}_2x(s_{k-2})$ which is held constant at the actuator (ZOH) until the next measurement is received.

Question 3: (8p)

1. (4p) Derive the exact discrete-time model $x_{k+1}^e = F_1(h, \tau)x_k^e + G_1(h, \tau)u_k^e$ capturing the dynamics of the described system at sampling instants. Note that you may need to consider extended states x^e and inputs u^e . Provide analytical expressions for $F_1(h, \tau)$ and $G_1(h, \tau)$, as a function of A , B , h and τ .
2. (2p) Find the ranges of values of h and τ for which the system is stable. Compare this controller implementation with the simpler controller from Question 2.
3. (2p) Let $\tau = h/2$. Design a new controller $\tilde{K}$, i.e. redesign $\bar{K}_1$ and $\bar{K}_2$, that extends the range of sampling intervals h for which the system remains stable in this case.

Finally, assume no delays $\tau = 0$, but consider that the system is implemented with a varying sampling interval taking two possible values h_1 and h_2 . When the sampling interval is h_1 the controller $\bar{K}$ is applied in sample-data fashion (zero order hold, constant input), and when the interval is h_2 the control $\bar{K}_2$ is applied in sample-data fashion (zero order hold, constant input).

Question 4: (6p)

1. (2p) Without assuming knowledge of the sequence of sampling intervals employed. Describe the set of LMIs that need to be solved to determine if such a controller guarantees stability for this NCS.
2. (2p) Assume now that the possible sequences of inter-sample intervals is restricted to be either¹: $(h_1h_2)^\omega$ or $(h_1)^\omega$. Can you simplify the check required to test stability of a given sampled-data controller in this case? e.g. can you reduce the size of the LMIs, if at all needed?
3. (2p) Under the same scenario as in the previous question. Find values of h_1 and h_2 guaranteeing the stability of the closed loop system. Verify the result through simulations.

¹The notation $(a)^\omega$ denotes the infinite repetition of that symbol, in the two cases in the question that means that the sequences end-up repeating forever the last inter-sample time